

# Ultra-Tiny VAD: Complete Build Plan

A from-scratch, commercially clean implementation of the kiloVAD paper

Based on: “VAD to the Bone: Ultra-Tiny Speech Activity Detection for Edge Deployment” (Analog Devices + UCLA, INTERSPEECH 2026)

Paper: <https://arxiv.org/abs/2607.25870>    Reference code/weights:  
<https://huggingface.co/spaces/kiloVAD-demo/kiloVAD>

**Version 2 — compiled September 1, 2026**

## What changed in v2 (read first)

**1. Dataset stack corrected.** The v1 plan recommended LibriVAD and Silero’s released labeled dataset. Both are **CC BY-NC-SA 4.0** — not usable for commercial training. They are replaced by a commercially clean stack (Section 1) plus a self-labeling recipe that uses the MIT-licensed Silero-VAD *model* to generate labels (Section 1.3). **2. Scope cuts.** The Optuna pruning-ratio search and INT4 QAT are cut: use the paper’s published pruning ratios and ship INT8. This removes most of the original weeks 4–5. **3. Timeline.** Now 2–3 weeks full-time with AI assistance, 4–5 weeks part-time (Section 10), sized to a \$30 Modal budget. **4. New:** full neural-network diagrams, per-phase analysis, a cross-VAD comparison table (Section 8), and an updated legality section (Section 11).

**Grounding note.** Architecture and training details are taken from the paper text and cross-checked against the authors’ released model code (`adivad_v2/utils/modules.py`, `class FlexibleVAD`; `preprocessing.py`; two released configs; checkpoints `best_model_epoch38.pth` and `pruned_model.pth`). Where the code is more specific than the paper prose, the code value is given and marked. The per-layer parameter count computed here is exactly 81,090, matching the released config, and the MAC count ( $\approx 1.66\text{M}$ ) matches the paper’s “1.7 M” — a built-in correctness check for the reimplement. License verdicts in Section 1 were re-verified against the source repos on September 1, 2026.

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# 1 Phase 0 — Legal footing and the dataset stack

Everything downstream depends on this being right, so it comes first. The build trains a *new* model from a paper description, on *commercially clean* data, and ships only self-trained weights.

## 1.1 Dataset license table (verified September 1, 2026)

| Dataset                                            | Size                 | License                                                                           | Commercial training                                                                                                                                                     | Source                                                                                                                                                          |
|----------------------------------------------------|----------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LibriSpeech train-clean-100                        | 100 h / 6.3 GB       | CC BY 4.0                                                                         | OK — keep attribution                                                                                                                                                   | <a href="https://openslr.org/12">https://openslr.org/12</a>                                                                                                     |
| Common Voice                                       | 3,000+ h (all langs) | CC0                                                                               | OK — public domain; download via Mozilla Data Collective, do not re-mirror                                                                                              | <a href="https://github.com/mozilla/legal-docs/blob/main/en/common_voice_terms.md">https://github.com/mozilla/legal-docs/blob/main/en/common_voice_terms.md</a> |
| MLS (Multilingual LibriSpeech)                     | ≈44k h total         | CC BY 4.0                                                                         | OK — keep attribution                                                                                                                                                   | <a href="https://openslr.org/94">https://openslr.org/94</a>                                                                                                     |
| VoxLingua107                                       | ≈6.6k h              | CC BY 4.0                                                                         | OK — keep attribution                                                                                                                                                   | <a href="https://cs.taltech.ee/staff/tanel.alumae/data/voxlangua107/">https://cs.taltech.ee/staff/tanel.alumae/data/voxlangua107/</a>                           |
| MUSAN (music/speech/noise)                         | ≈109 h               | CC BY 4.0                                                                         | OK — curators already excluded NC sources; keep the per-file license and attribution files                                                                              | <a href="https://openslr.org/17">https://openslr.org/17</a>                                                                                                     |
| DNS Challenge noise (Interspeech 2020)             | tens of GB           | mixed: AudioSet CC BY 4.0; Freesound CC0-only (already filtered); DEMAND CC BY-SA | OK with conditions — attribution; DEMAND share-alike binds adaptations of the clips, standard reading is it does not reach trained weights; drop DEMAND if conservative | <a href="https://github.com/microsoft/DNS-Challenge">https://github.com/microsoft/DNS-Challenge</a>                                                             |
| Wind noise simulator (Mirabilli et al. 2022)       | code                 | check repo before use                                                             | Generator code, not data — verify its license; alternative: record/synthesize own wind                                                                                  | paper ref.                                                                                                                                                      |
| Silero-VAD <i>model</i> (labeling tool + baseline) | —                    | MIT                                                                               | OK — model outputs are not encumbered; using it to label your own data is fine                                                                                          | <a href="https://github.com/snakers4/silero-vad">https://github.com/snakers4/silero-vad</a>                                                                     |
| Silero released <i>labeled dataset</i>             | —                    | CC BY-NC-SA 4.0                                                                   | NOT OK for commercial training                                                                                                                                          | <a href="https://github.com/snakers4/silero-vad/blob/master/datasets/README.md">https://github.com/snakers4/silero-vad/blob/master/datasets/README.md</a>       |
| LibriVAD                                           | 15 GB tier etc.      | CC BY-NC-SA 4.0                                                                   | NOT OK for commercial training (v1 plan error, corrected)                                                                                                               | <a href="https://huggingface.co/datasets/LibriVAD/LibriVAD">https://huggingface.co/datasets/LibriVAD/LibriVAD</a>                                               |

|                                 |   |              |                                               |                                                                                                                     |
|---------------------------------|---|--------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| kiloVAD released code + weights | — | CC BY-NC 4.0 | NOT OK to ship — use as parity reference only | <a href="https://huggingface.co/spaces/kiloVAD-demo/kiloVAD">https://huggingface.co/spaces/kiloVAD-demo/kiloVAD</a> |
|---------------------------------|---|--------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------|

## 1.2 Why the exclusions do not hurt

Silero’s labeled dataset is mostly packaging: its permissive components (Common Voice CC0, VoxLingua107 CC BY, MLS CC BY) can be pulled from the original sources and labeled with Silero-VAD itself, giving the same coverage with no NC problem. LibriVAD is LibriSpeech plus noise mixed at known SNRs — the same thing this plan generates directly, so the generated dataset is yours, not theirs.

## 1.3 Self-labeling recipe (replaces the MFA/LibriVAD labeling path)

The paper labels LibriSpeech with Montreal Forced Aligner. We replace that with Silero-VAD self-labeling, which works on any corpus, needs no Kaldi, and is commercially clean end to end.

- Run Silero-VAD over every speech clip.** 16 kHz input, 512-sample (32 ms) chunks, hidden state carried forward across the clip (never reset mid-clip). Collect the per-chunk speech probability.
- Threshold with hysteresis.** Speech starts when  $\text{prob} \geq 0.5$  and ends when  $\text{prob} < 0.35$  (Silero’s own defaults: threshold 0.5,  $\text{neg\_threshold} = \text{threshold} - 0.15$ ). Enforce minimum speech duration 250 ms, minimum silence 100 ms, pad speech segments by 30 ms — the `get_speech_timestamps` defaults. This suppresses label flicker.
- Rasterize to the 10 ms frame grid.** A frame is speech if its center falls inside a speech segment. This yields one label per 10 ms hop, exactly matching the model’s frame rate.
- Quality check.** (a) Listen to 50–100 random clips with the label track overlaid on the spectrogram — boundary errors are audible. (b) Where forced alignments already exist (e.g., LibriSpeech MFA alignments from Lugosch et al. 2019), measure boundary disagreement; median offset under  $\approx 40$  ms is fine for training with label smoothing. (c) Check the speech-fraction histogram per corpus; a corpus that is 95%+ “speech” gives the model almost no negative examples — fix by appending synthetic silences/pauses.
- Label-noise expectation.** Silero labels are model-generated and slightly softer at boundaries than MFA phoneme alignments. The training recipe already includes label smoothing  $\varepsilon = 0.09$ , which absorbs this. Validate empirically: the Phase 4 parity gate (AUC within the paper’s band) will catch a labeling problem.
- Mix noise at controlled SNR.** Per training sample: 25% clean speech; 25% mixed with synthetic wind at  $-5$  dB SNR; 50% mixed with DNS/MUSAN noise at SNR drawn uniformly from  $\{-10, -5, 0, +5, +10\}$  dB. Half of the noise-mixed samples also get simulated room reverb (py-roomacoustics RIRs). No pure-noise-only samples (matches the released config `libri_dns_full_no_pure_noi`).

## 1.4 Speech-mix strategy: two runs

**Run A (parity run, week 1–2):** LibriSpeech train-clean-100 only, exactly the paper’s data mix. Purpose: reproduce 0.862 AUC and prove the pipeline is correct against the official weights. **Run B (product run):** LibriSpeech + Common Voice + MLS + VoxLingua107 (suggest 40/25/20/15 of speech hours) for

multilingual robustness — the property a product VAD actually needs and where your model can beat the paper’s numbers. Run B only starts after Run A passes its gate.

| Phase facts               |                                                                                                                                  |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Effort (full-time)</b> | 1 day (decisions) + download time                                                                                                |
| <b>Effort (part-time)</b> | same                                                                                                                             |
| <b>Exit criteria</b>      | License table confirmed; corpora downloaded; Silero labeling script runs end-to-end on 10 clips; QC spot-check done.             |
| <b>Failure points</b>     | Missing per-file attribution on MUSAN; accidentally pulling LibriVAD out of habit; Common Voice download via unofficial mirrors. |

## 2 Phase 1 — Audio frontend

### 2.1 What happens

All audio is 16 kHz mono. The frontend converts a 200 ms waveform segment (3,200 samples) into a normalized log-mel tensor of shape (batch, 64, 21). Steps, in order (from `preprocessing.py` and paper Section 2.1):

1. **Pre-emphasis:**  $y[n] = x[n] - 0.97 x[n - 1]$  (non-trainable).
2. **Waveform normalization:** subtract mean, divide by std over the input window. The paper’s “amplitude-agnostic” preprocessing: forces the model to learn relative spectral patterns, not absolute energy — robustness across mics and gains.
3. **Mel spectrogram** (`torchaudio.MelSpectrogram`): `n_fft` 512, `win_length` 400 (25 ms Hann, zero-padded to 512), `hop` 160 (10 ms),  $f_{\min}$  0,  $f_{\max}$  8000, 64 mels, `power` 2.0, `center=True` (reflect pad 256 each side).
4. **Log compression:**  $\log(\text{mel} + 10^{-5})$ .
5. **Per-bin window normalization:** for each of the 64 mel bins, subtract the mean and divide by the std across the 21 frames ( $\text{eps } 10^{-5}$ ).

| Parameter                                                | Value                           | Notes                                       |
|----------------------------------------------------------|---------------------------------|---------------------------------------------|
| Sample rate                                              | 16,000 Hz                       | mono                                        |
| Input context ( <code>frame_sec</code> )                 | 0.2 s (200 ms)                  | detection latency; multiple of 10 ms        |
| Output time frames                                       | 21                              | $3200/160 + 1$ ( <code>center=True</code> ) |
| <code>n_fft</code> / <code>win</code> / <code>hop</code> | 512 / 400 / 160                 | Hann; 400-sample window padded to 512       |
| Mel bins                                                 | 64 (32 supported, $-0.003$ AUC) | 0–8000 Hz                                   |
| Model input tensor                                       | (batch, 64, 21)                 | channels = mel bins                         |

**Gotcha:** `frame_sec` must be an exact multiple of the 10 ms hop and at least 30 ms, or frames are wasted / padding breaks (the released code raises for this). 200 ms → exactly 21 frames.

| Phase facts               |                                                                                                                                                                    |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Effort (full-time)</b> | 0.5 day                                                                                                                                                            |
| <b>Effort (part-time)</b> | 1 day                                                                                                                                                              |
| <b>Exit criteria</b>      | Feature tensors match the official HF preprocessing numerically on fixed audio inputs.                                                                             |
| <b>Failure points</b>     | Normalization order (waveform z-norm <i>before</i> mel; per-bin norm <i>after</i> log-mel) — swapping silently breaks parity; frame-count off-by-one (21, not 20). |

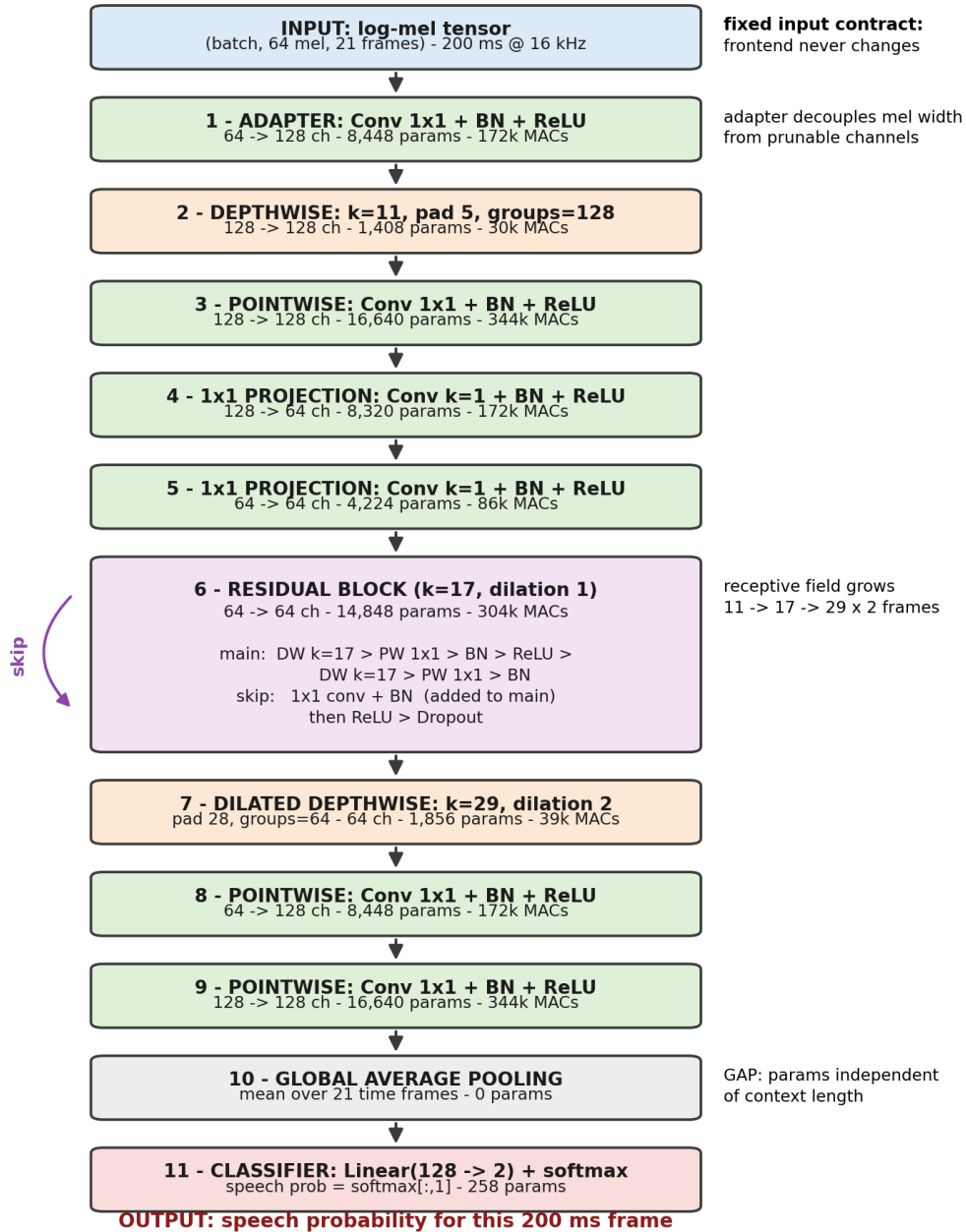
## 3 Phase 2 — Model architecture

### 3.1 What happens

Paper summary (Section 2.1): “an adapter layer ( $n_{\text{mels}} \rightarrow 128$ ), a depthwise separable block (temporal kernel 11), two  $1 \times 1$  projection layers ( $128 \rightarrow 64 \rightarrow 64$  channels), a residual block (kernel 17), a dilated block (kernel 29, dilation 2), a pointwise conv, global average pooling, and a binary classifier.” The official code implements exactly this with Conv1d over (channels, time). All convs are bias-free; every conv is followed by BatchNorm1d (eps  $10^{-3}$ , momentum 0.1, per code); activations are ReLU; stride 1 with same padding throughout, so the time dimension stays 21 until pooling.

**FlexibleVAD / kiloVAD architecture - 81,090 params, ~1.66M MACs per 200 ms decision**

All convs bias-free; every conv followed by BatchNorm1d (eps 1e-3) + ReLU; stride 1, same padding (time stays 21 until GAP)

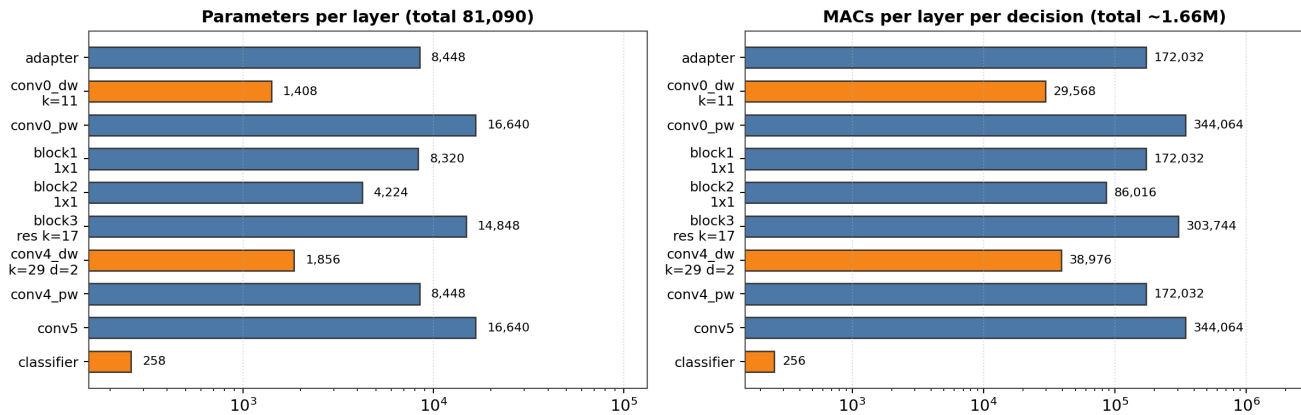


### 3.2 Layer-by-layer parameter and MAC table

| #            | Layer (code name)          | In → out  | Params        | MACs          | Type                    |
|--------------|----------------------------|-----------|---------------|---------------|-------------------------|
| 1            | adapter_conv + BN          | 64 → 128  | 8,448         | 172,032       | 1×1 conv                |
| 2            | conv0_dw (k=11, g=128)     | 128 → 128 | 1,408         | 29,568        | depthwise               |
| 3            | conv0_pw + BN              | 128 → 128 | 16,640        | 344,064       | pointwise               |
| 4            | block1 (1×1 proj.)         | 128 → 64  | 8,320         | 172,032       | pointwise               |
| 5            | block2 (1×1 proj.)         | 64 → 64   | 4,224         | 86,016        | pointwise               |
| 6            | block3 (residual, k=17)    | 64 → 64   | 14,848        | 303,744       | DW-sep. residual        |
| 7            | conv4_dw (k=29, d=2, g=64) | 64 → 64   | 1,856         | 38,976        | dilated depthwise       |
| 8            | conv4_pw + BN              | 64 → 128  | 8,448         | 172,032       | pointwise               |
| 9            | conv5 + BN                 | 128 → 128 | 16,640        | 344,064       | pointwise               |
| 10           | global average pooling     | 128       | 0             | 2,688         | pooling                 |
| 11           | classifier Linear(128→2)   | 128 → 2   | 258           | 256           | linear                  |
| <b>Total</b> |                            |           | <b>81,090</b> | <b>≈1.66M</b> | matches released config |

Params include BN affine weights (2 per channel); convs are bias-free; MACs use 21 output frames. **block3 expanded:** (1) DW k=17 pad 8 → PW 1×1 → BN → ReLU → Dropout; (2) DW k=17 → PW 1×1 → BN (no activation); (3) skip branch: 1×1 conv + BN added to main; (4) ReLU → Dropout.

Where the budget goes - the two dense 128x128 pointwise convs dominate both params and MACs (prune them first)



### 3.3 Design rules baked into the architecture

- **CNN-only:** every layer except the final linear is conv, BN, ReLU, or pooling. No Sinc filters, no GRUs, no novel activations. Static graph, TFLM-compatible.
- **1×1 adapter:** decouples the fixed 64-bin mel input from internal widths, so structured pruning of internal channels never touches the input interface.
- **Global average pooling instead of flatten:** parameter count is independent of input length — the same weights run at 60–360 ms context with no architectural change.

- **Binary head:** Linear(128  $\rightarrow$  2); training returns raw logits, inference returns the class-1 (speech) softmax probability.
- **Receptive field:** kernels 11  $\rightarrow$  17  $\rightarrow$  29 (dilation 2) progressively expand temporal context over the 21-frame window; 200 ms  $\approx$  one syllable at the 4–5 Hz syllable rate (the paper’s stated justification for the context length).

| Phase facts               |                                                                                                                                                                                   |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Effort (full-time)</b> | 1 day                                                                                                                                                                             |
| <b>Effort (part-time)</b> | 2–3 days                                                                                                                                                                          |
| <b>Exit criteria</b>      | (1) Param count exactly 81,090. (2) Official checkpoint loads with <code>strict=True</code> . (3) Logits match the official model to float tolerance on fixed inputs.             |
| <b>Failure points</b>     | BN eps $10^{-3}$ not PyTorch default $10^{-5}$ (shows up as a systematic logit offset); depthwise/pointwise pairing; forgetting the residual block’s second BN has no activation. |

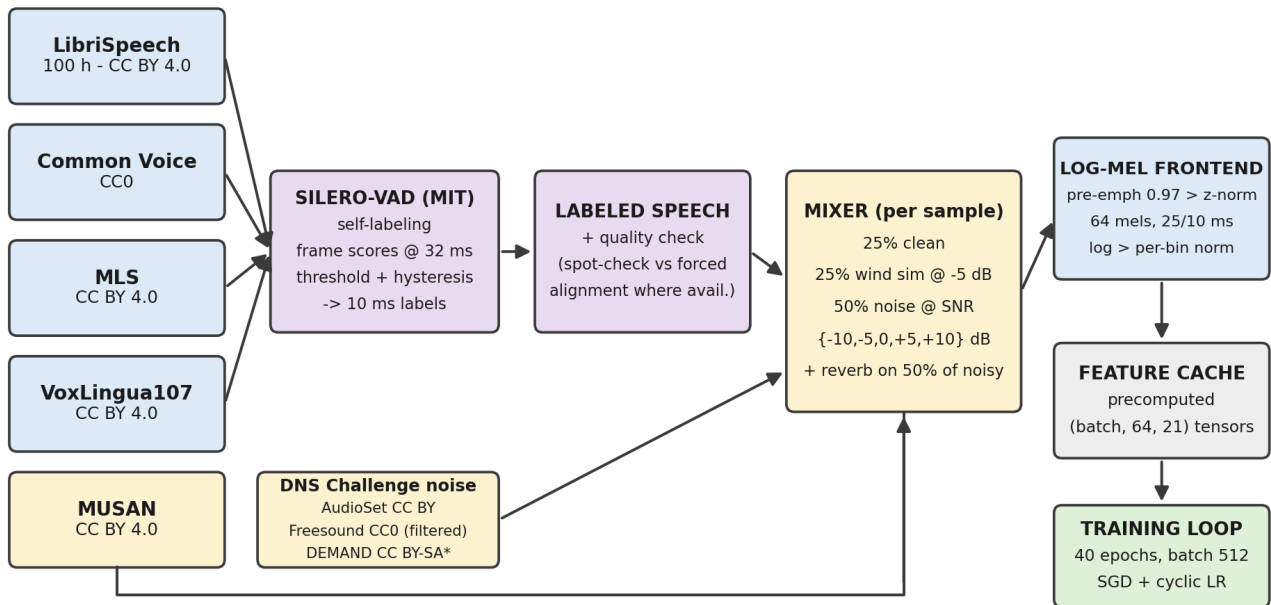
## 4 Phase 3 — Data pipeline build

### 4.1 What happens

Build the corpus once, cache features, never touch it again. All speech is labeled with the Section 1.3 recipe, mixed with noise at controlled SNRs, reverbed, then pushed through the Phase 1 frontend and cached as tensors.

#### Training-data pipeline - commercially clean stack (v2, license-corrected)

\*DEMAND share-alike binds adaptations of the clips, not the trained model (standard reading). Excluded as CC BY-NC-SA: Silero labeled dataset, LibriVAD.



### 4.2 Mixing ratios (paper-faithful)

| Component          | Share        | Detail                                                                              |
|--------------------|--------------|-------------------------------------------------------------------------------------|
| Clean speech       | 25%          | labeled speech, no noise                                                            |
| Wind-mixed         | 25%          | synthetic wind (Mirabilii et al. 2022 airflow simulator) at $-5$ dB SNR             |
| Noise-mixed        | 50%          | DNS Challenge + MUSAN noise at $\text{SNR} \in \{-10, -5, 0, +5, +10\}$ dB, uniform |
| Reverberation      | 50% of noisy | simulated RIRs (pyroomacoustics / gpuRIR)                                           |
| Pure-noise samples | 0%           | released config name: <code>libri_dns_full_no_pure_noise_v2</code>                  |

**Cache:** features are precomputed and cached (released config: `use_precomputed: true`); the cache, not the audio, is what training reads. Disk: tens of GB for DNS noise,  $\approx 6.3$  GB LibriSpeech, plus the feature cache.

| Phase facts        |                                                                                                                                                                               |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Effort (full-time) | 3–4 days (mostly downloads, labeling pass, cache build)                                                                                                                       |
| Effort (part-time) | 7–8 days                                                                                                                                                                      |
| Exit criteria      | Feature cache built; label QC report done (Section 1.3 step 4); five eval categories materialized: Clean, Windy, +DNS Synthetic, DNS Speech+Noise, DNS Pure Noise (held out). |
| Failure points     | SNR gain math errors (normalize speech and noise levels before scaling); label rasterization boundary rounding; running out of disk mid-cache.                                |

## 5 Phase 4 — Base training

### 5.1 Recipe

| Hyperparameter  | Value                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------|
| Optimizer       | SGD, momentum 0.9, Nesterov                                                                        |
| Weight decay    | $8.75 \times 10^{-4}$                                                                              |
| LR schedule     | cyclic: 4-epoch warmup to $3.5 \times 10^{-3}$ , 16-epoch hold, 20-epoch cosine decay to $10^{-5}$ |
| Loss            | cross-entropy, label smoothing $\varepsilon = 0.09$                                                |
| Epochs / batch  | 40 / 512                                                                                           |
| Model selection | best checkpoint (released: <code>best_model_epoch38.pth</code> )                                   |
| Replicates      | paper: 10 seeds, mean $\pm$ 95% CI. <b>Budget variant: 3 seeds</b> (see cost note)                 |

**Expected result:**  $0.862 \pm 0.001$  AUC and  $0.796 \pm 0.001$  frame F1 on AVA-Speech (causal, 200 ms, 64 mels, paper protocol, 10 seeds). With Silero-labeled data and 3 seeds, treat “within  $\approx 0.005$  of 0.862” as a pass — labeling noise and fewer seeds widen the band.

### 5.2 Modal cost plan (\$30 budget)

| Item                            | Est. cost               | Notes                                                                     |
|---------------------------------|-------------------------|---------------------------------------------------------------------------|
| Labeling pass (CPU)             | \$1–2                   | Silero on CPU is $<1$ ms per 32 ms chunk; hours of audio label in minutes |
| Base training Run A, 3 seeds    | \$6–15                  | T4 ( $\approx 0.60$ /h), 2–4 h per 40-epoch seed on cached features       |
| Distillation fine-tunes (8 ep.) | \$1–2                   | per pruned configuration                                                  |
| Eval + export runs              | \$1–2                   | CPU is fine                                                               |
| Run B (multilingual, optional)  | \$5–10                  | only after Run A passes                                                   |
| <b>Total (A only / A+B)</b>     | <b>\$9–19 / \$14–29</b> | headroom for mistakes inside \$30                                         |

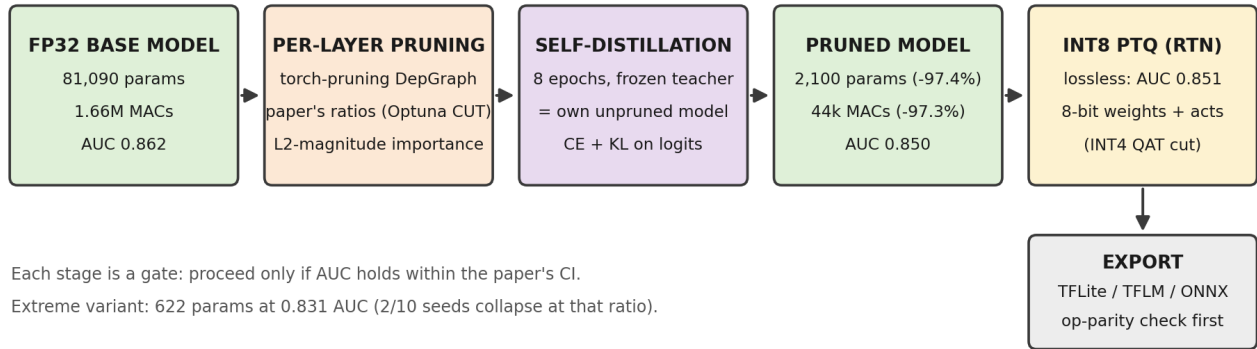
**Flag:** if the Fliko music endpoint still runs on Modal it eats the same credits — check the balance before starting.

| Phase facts               |                                                                                                                                                                                                 |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Effort (full-time)</b> | 2–4 days (mostly GPU wait)                                                                                                                                                                      |
| <b>Effort (part-time)</b> | 5 days                                                                                                                                                                                          |
| <b>Exit criteria</b>      | Run A mean AUC within $\approx 0.005$ of the paper’s 0.862 on the causal AVA-Speech protocol.                                                                                                   |
| <b>Failure points</b>     | Cyclic-LR mis-implementation (warmup 4 / hold 16 / decay 20); dataloader starving the GPU (12 workers, prefetch 4, pin_memory); training before the Phase 2 parity gate passes — bugs compound. |

## 6 Phase 5 — Pruning and self-distillation

### 6.1 What happens

**Compression pipeline - prune, distill, quantize (paper numbers, causal 200 ms AVA-Speech)**



1. **Tooling:** torch-pruning (DepGraph). The dependency graph prunes coupled channels together (depthwise/pointwise pairs, residual adds), preserving valid shapes.
2. **Importance:** L2-norm of each channel's weights; remove the least-contributing channels.
3. **Per-layer ratios, not global:** the paper shows uniform global pruning collapses below 2k params (layer collapse); per-layer ratios stay stable down to 622 params. **Scope cut:** skip the Optuna multi-objective search entirely — hardcode the paper's published ratio vector for the 2.1k configuration. This deletes the single biggest compute item in the v1 plan ( $\approx 1$  GPU-day).
4. **Self-distillation:** the unpruned trained model is the teacher (frozen); fine-tune the pruned student 8 epochs with CE (same label smoothing) + KL divergence between student and teacher logits. No separate teacher needed.
5. **Transfer (optional):** the paper optimizes ratios on one seed and transfers to 10 models; at the most aggressive point 2/10 seeds collapse. With 3 seeds, prune each seed's own model with the same ratio vector.

| Model                            | Params            | MACs/infer. | AVA-Speech AUC (causal, 200 ms)           |
|----------------------------------|-------------------|-------------|-------------------------------------------|
| Unpruned baseline                | 81.1 k            | 1.7 M       | $0.862 \pm 0.001$                         |
| Per-layer pruned + KD (featured) | 2.1 k<br>(-97.4%) | 44 k        | $0.850 \pm 0.007$ (F1 $0.783 \pm 0.004$ ) |
| Per-layer pruned + KD (extreme)  | 622               | —           | 0.831                                     |

| Phase facts        |                                                                                                                                      |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Effort (full-time) | 2 days                                                                                                                               |
| Effort (part-time) | 3 days                                                                                                                               |
| Exit criteria      | 2.1k model at $\approx 0.850$ AUC (within CI); pruned graph passes op-parity vs. its fp32 parent.                                    |
| Failure points     | Pruning pointwise without depthwise/residual partners (DepGraph prevents this); uniform ratios; distilling with an unfrozen teacher. |

## 7 Phase 6 — Quantization

### 7.1 INT8 PTQ (the ship target)

Post-training INT8 with round-to-nearest is **lossless** in the paper: the 10k model stays at 0.861 AUC and the 2.1k model at 0.851, both matching FP32. Do this first and ship this. 8-bit weights and activations run on every embedded target.

### 7.2 INT4 angle-aware QAT (cut from this build; documented for completeness)

The paper’s contribution: freeze the FP32 classifier as fixed class prototypes, quantize only backbone weights, and train with an align-repel cosine loss  $L_i = [1 - \cos(f_i, w_{y_i}^{FP})] + \lambda \max(0, \max_{c \neq y_i} \cos(f_i, w_c^{FP}))$ , annealed soft-to-hard. Result: INT4 STE QAT collapses on small models (0.693 AUC at 2.1k) while angle-aware recovers part of it (0.719) — still a 1–4% relative gain, not a fix. **Verdict:** INT4 remains a lossy regime; not worth the build time when INT8 is free and lossless. Revisit only if a target MCU cannot fit  $\approx 2.1$ k INT8 weights ( $\approx 2.1$  KB).

| Phase facts        |                                                                                                                |
|--------------------|----------------------------------------------------------------------------------------------------------------|
| Effort (full-time) | 1 day                                                                                                          |
| Effort (part-time) | 2 days                                                                                                         |
| Exit criteria      | INT8 model AUC = FP32 AUC within noise; TFLite/ONNX exports match fp32 outputs op-for-op on fixed inputs.      |
| Failure points     | Quantizing BN-folded graphs incorrectly; export before op-parity checks; attempting INT4 “because it’s there”. |

## 8 Phase 7 — Evaluation and the cross-VAD comparison

### 8.1 Causal protocol (non-negotiable)

Reported VAD numbers are frequently inflated by non-causal sliding-window inference with 87.5% overlap (AtomicVAD: 0.903 non-causal vs. 0.869 causal on AVA-Speech). Every number you report must come from the strictly causal per-frame protocol, and the protocol must be stated next to every number:

1. Stream each test clip in non-overlapping 200 ms hops; one speech probability per hop from audio up to and including that hop only — no future context, no overlap, no median filter, no hangover.
2. Frame labels: bin AVA-Speech segment annotations to the 10 ms grid (a frame is speech if it overlaps any annotated segment); aggregate to the 200 ms grid by majority.
3. Sweep the decision threshold for the ROC. Report AUC, best F1 (with its threshold), and FPR at TPR = 0.95 — the operating point that matters for always-on triggers.
4. Silero baseline: native 512-sample (32 ms) chunks, state carried causally, chunk scores mapped onto the 200 ms grid (average chunk scores whose centers fall in the frame); record the mapping in the report.

### 8.2 Benchmark sets

| Set                                  | Role                                                                                                                                      |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| AVA-Speech                           | primary benchmark; out-of-domain vs. training data, tests generalization; expect link rot on some YouTube clips — document recovery count |
| Held-out Lib-riSpeech/DNS categories | in-domain sanity: Clean, Windy, +DNS Synthetic, DNS Speech+Noise, DNS Pure Noise (pure noise should score near-zero speech prob)          |
| FLEURS-VAD-102 (optional)            | multilingual; lets you compare on the same ground as the FireRedVAD report's published table below                                        |

### 8.3 Cross-VAD comparison (published numbers)

Two benchmark families appear below. **AVA-Speech causal** numbers come from the kiloVAD paper (strictly causal per-frame). **FLEURS-VAD-102** numbers come from the FireRedASR2S tech report (arXiv:2603.10420; frame-level, fixed 0.5 threshold, 9,443 files across 102 languages). Do not compare numbers across the two families directly — protocols and data differ. “—” = not published in the cited source.

| Model                | Params | Footprint                 | Compute             | Accuracy (as published)         | License / notes                                 |
|----------------------|--------|---------------------------|---------------------|---------------------------------|-------------------------------------------------|
| kiloVAD full (paper) | 81.1 k | 324 KB fp32 / 81 KB INT8  | 1.66M MACs / 200 ms | AVA causal: 0.862 AUC, 0.796 F1 | weights CC BY-NC 4.0 — reimplement, do not ship |
| kiloVAD pruned + KD  | 2.1 k  | 8.4 KB fp32 / 2.1 KB INT8 | 44k MACs / 200 ms   | AVA causal: 0.850 AUC, 0.783 F1 | same                                            |

|                                 |           |                          |                                                   |                                                                               |                                                            |
|---------------------------------|-----------|--------------------------|---------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------|
| kiloVAD ex-treme                | 622       | ≈2.5 KB fp32             | —                                                 | AVA causal: 0.831 AUC                                                         | 2/10 seeds collapse at this ratio                          |
| Silero-VAD (v5/v6)              | ≈545.6 k  | ≈2.2 MB fp32 (JIT ≈2 MB) | <1 ms per 32 ms chunk, 1 CPU thread (repo README) | FLEURS-102: 97.99% AUC, 95.95% F1                                             | MIT; stateful (2×128 LSTM state), 32 ms chunks             |
| FireRedVAD                      | ≈0.6 M    | —                        | —                                                 | FLEURS-102: 99.60% AUC, 97.57% F1, FAR 2.69%                                  | DFSMN; streaming + mVAD; repo: Apache 2.0                  |
| TEN-VAD                         | —         | —                        | —                                                 | FLEURS-102: 97.81% AUC, 95.19% F1                                             | —                                                          |
| FunASR-VAD                      | —         | —                        | —                                                 | FLEURS-102: 90.91% F1 (AUC n/a — no posterior)                                | —                                                          |
| WebRTC VAD                      | n/a (GMM) | tens of KB               | negligible                                        | FLEURS-102: 52.30% F1 (AUC n/a)                                               | BSD; the “good enough” baseline everyone ships             |
| FDE-Mamba (2026)                | 145.5 k   | —                        | +62.7% kFLOPs/frame vs FDE-RNN; 3.16× better RTF  | <i>Different task:</i> personal VAD mAP 0.9605, acc 89.87% (LibriSpeech PVAD) | Mamba/SSM — not TFLM-friendly; not comparable to the above |
| MarbleNet / TinyVAD / AtomicVAD | —         | —                        | —                                                 | AtomicVAD AVA: 0.903 non-causal / 0.869 causal; 630 ms context                | from kiloVAD paper Table 1; params not restated there      |

**How to read this:** on raw AUC you will not beat Silero or FireRedVAD out of the gate — they train on vastly more (and, for FireRed, human-labeled) data. The winning frame is **efficiency per parameter**: 2.1k params and 44k MACs is ≈250× smaller and orders of magnitude cheaper than anything else on the table, at 0.85 causal AUC. The README benchmark table should be: params, size, latency, MACs, INT8 drop, accuracy-per-param. On that table this model beats everyone, including the original paper’s framing.

## 8.4 Metric definitions

| Metric         | How to measure                                                                                                                                                                            |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frame AUC      | ROC area over per-frame speech probabilities (primary)                                                                                                                                    |
| Best frame F1  | max F1 over threshold sweep; report the threshold                                                                                                                                         |
| FPR @ TPR=0.95 | the paper's pruning objective; the always-on operating point                                                                                                                              |
| Latency        | (a) algorithmic = input context (200 ms); (b) wall-clock per decision, 1 thread, mean + p99 over $\geq 10k$ decisions                                                                     |
| Memory         | model footprint (fp32 + INT8) + peak RSS during streaming, frontend included                                                                                                              |
| RTF            | CPU time per second of audio: server CPU, ARM Cortex-A board, optionally TFLM on Cortex-M (kiloVAD only — Silero's stateful LSTM complicates TFLM; that gap is itself a headline finding) |

| Phase facts        |                                                                                                                                                                                       |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Effort (full-time) | 2–3 days                                                                                                                                                                              |
| Effort (part-time) | 3 days                                                                                                                                                                                |
| Exit criteria      | Full comparison table filled with CIs and hardware numbers; protocol disclosed next to every number; official-weights metric parity ( $\approx 0.862$ / $\approx 0.850$ ) reproduced. |
| Failure points     | Any overlap/smoothing in eval (invalidates everything); comparing across benchmark families; trusting repo-README latency numbers instead of measuring.                               |

## 9 Phase 8 — Export and hardware benchmark

The graph is CNN-only and static, so it converts to TFLite/TFLM and ONNX with standard ops only. Verify op-for-op export parity on a fixed input before benchmarking. Fix one CPU thread for all latency/RTF numbers; record CPU model, clock, OS, and runtime versions. Report kiloVAD fp32, kiloVAD INT8, and Silero (PyTorch + ONNX Runtime) per-decision wall time on the same machine; then an ARM Cortex-A board (Raspberry Pi 4/5); then TFLM on a Cortex-M for the 2.1k INT8 model as the stretch goal.

| Phase facts        |                                                                                                                                                         |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Effort (full-time) | 1–2 days                                                                                                                                                |
| Effort (part-time) | 2–3 days                                                                                                                                                |
| Exit criteria      | Exported graphs bit-match fp32 within tolerance; hardware table filled; TFLM build runs on the Cortex-M target (stretch).                               |
| Failure points     | TFLite op coverage surprises (there should be none — conv/BN/ReLU/GAP only); benchmarking before parity; batch-size $> 1$ sneaking into latency claims. |

## 10 Timeline



### 10.1 Full-time with AI assistance (~3 weeks)

| Days  | Phase                   | Exit gate                                 |
|-------|-------------------------|-------------------------------------------|
| 1–4   | P1 data + self-labeling | cache built, QC done                      |
| 3–5   | P2 frontend + model     | 81,090 params; official-weight parity     |
| 6–9   | P3 base training        | Run A AUC within $\approx 0.005$ of 0.862 |
| 10–11 | P4 prune + distill      | 2.1k at $\approx 0.850$                   |
| 12    | P5 INT8 + export        | lossless INT8; op-parity                  |
| 13–15 | P6 eval + benchmark     | comparison table + hardware numbers       |

### 10.2 Part-time (~5 weeks)

Same gates, roughly doubled phase lengths: P1 days 1–8, P2 days 6–12, P3 days 13–17, P4 days 18–20, P5 days 21–22, P6 days 23–25. The v1 paper-rigor plan (10 seeds, Optuna search, INT4 QAT) was 6 weeks *full-time* — the cuts, not heroics, are what make 2–3 weeks realistic.

**LOC estimate:** frontend  $\approx 200$ , model  $\approx 250$ , data pipeline  $\approx 500$ , training + eval  $\approx 500$ , prune/distill  $\approx 300$ , quantize/export  $\approx 250$  — about 2k lines of real code, 3k with tests. AI writes the model/export/eval boilerplate fast; it cannot speed up numeric-parity debugging or GPU wait.

## 11 Legality (updated September 1, 2026)

Verdict: **GO, with conditions.** Not legal advice; run a real FTO review before a paid launch.

1. **Write your own code from the paper.** ADI's released code and weights are CC BY-NC 4.0 — research use only. Use them as a parity reference; never fold them into the product. Papers describe ideas; ideas are not copyrighted, their expression (the code) is.
2. **Train and ship only your own weights**, on the Section 1 clean stack.
3. **Do not name the product “kiloVAD”** — that is their name/mark. Pick your own.
4. **Attribution:** credit the CC BY datasets (LibriSpeech, MLS, VoxLingua107, MUSAN) in an ATTRIBUTION file; keep MUSAN's per-file LICENSE files intact in any redistribution of the data itself.
5. **DEMAND caveat:** CC BY-SA clips inside DNS noise — share-alike binds adaptations of the clips; the standard reading is that trained model weights are not an adaptation of the clips. If you want zero ambiguity, drop DEMAND; DNS has plenty of CC BY/CC0 noise.
6. **Silero:** model is MIT (labeling tool + benchmark baseline: fine). Their labeled dataset is CC BY-NC-SA (do not touch). Same for LibriVAD.
7. **Patent blind window:** the paper is from Analog Devices; patent applications publish  $\approx 18$  months after filing, so there may be pending claims nobody can see yet. The architecture (depthwise-separable CNN + GAP) is generic and likely safe; the specific angle-aware QAT and pruning-ratio search are the more claimable pieces — and both are cut from this build anyway.

## 12 Failure points checklist (carry into the build)

1. Frame-count off by one: with center=True, frames = samples/hop + 1 = 21.
2. Non-multiple frame\_sec: window must be an exact multiple of the 10 ms hop.
3. Normalization order: pre-emphasis  $\rightarrow$  waveform z-norm  $\rightarrow$  mel  $\rightarrow$  log  $\rightarrow$  per-bin norm. Swaps silently break parity.
4. BN hyperparameters: eps  $10^{-3}$  (not the  $10^{-5}$  default), momentum 0.1.
5. Evaluating with overlap or smoothing: inflates AUC, invalidates the comparison.
6. Uniform pruning ratios: layer collapse below  $\approx 2k$  params.
7. Pruning grouped channels inconsistently: always through DepGraph.
8. QAT through the classifier (if INT4 is ever attempted): the FP classifier stays frozen.
9. Seed variance: the pruned model's CI is  $\pm 0.007$  AUC — single-seed numbers are noise; report 3 seeds minimum, 10 for publication-grade.
10. Label extraction: sloppy 10 ms rasterization adds label noise that looks like model error.
11. License drift: someone pulls LibriVAD or the Silero labeled dataset out of habit — the Section 1 table is the authority.

## 13 Sources

- Paper: <https://arxiv.org/abs/2607.25870> (HTML: <https://arxiv.org/html/2607.25870v1>)
- Official weights/code/demo (CC BY-NC 4.0): <https://huggingface.co/spaces/kiloVAD-demo/kiloVAD>
- Silero-VAD (MIT): <https://github.com/snakers4/silero-vad>; labeled dataset terms (CC BY-NC-SA): <https://github.com/snakers4/silero-vad/blob/master/datasets/README.md>
- LibriVAD (CC BY-NC-SA, excluded): <https://huggingface.co/datasets/LibriVAD/LibriVAD>
- Common Voice terms (CC0): [https://github.com/mozilla/legal-docs/blob/main/en/common\\_voice\\_terms.md](https://github.com/mozilla/legal-docs/blob/main/en/common_voice_terms.md)
- LibriSpeech: <https://openslr.org/12>    MLS: <https://openslr.org/94>    MUSAN: <https://openslr.org/17> (corpus paper: <https://arxiv.org/html/1510.08484>)
- DNS Challenge: <https://github.com/microsoft/DNS-Challenge>
- FireRedVAD / FLEURS-VAD-102 numbers: FireRedASR2S tech report, <https://arxiv.org/abs/2603.10420v1>
- FDE-Mamba: <https://www.mdpi.com/2076-3417/16/10/4688>
- AVA-Speech, torch-pruning (CVPR 2023), Montreal Forced Aligner, Lugosch et al. 2019 alignments, pyroomacoustics